

WATER MANAGEMENT can't be the preserve of government alone. It has to be solved with the help of the community and private players.

They call it the elixir of life but with changing times, water has also become a driving force in economic activities, traditionally in agriculture, and now increasingly in the industry. No wonder, there has been a tendency to take for granted the availability of water and show scant responsibility to the understanding of the ecological processes that make it available to us.

The freshwater sources that are needed to meet human needs and are easily accessible (as in lakes, rivers and shallow groundwater aquifers) constitute less than 0.3 per cent of the total water available on the planet. The availability is highly skewed on the terrestrial surface. India alone exhausts about 1,000 cubic metres of water per capita per year. With this amount, we have to make do with the satisfaction of domestic needs, food production, industrial activities, pollution control and all the environmental needs of other water systems. Any further decline would be taken as an indication of serious water scarcity. Add to that the acute inequity in the country in access to water, that the national level averages conveniently make invisible.

A recent report of the United Nations Development Programme (UNDP) observes, "Ever since Independence, poor Indians have been promised free, safe, household water." Even after 61 years of Independence, more than 300 million Indians do not have access to safe drinking water. Even when they have, as the UNDP report says, poor people "still have to pay for their water, sometimes 10 to 20 times more than their richer neighbours".

One of the effective ways of

dealing with the problem is to have a fundamentally holistic and inclusive water systems management. This new approach has to move beyond the confines of traditional engineering, which had played a successful role in the era of reinforced concrete structures, starting with the construction of the Hoover dam in the USA. After decades of following that strategy based on structural interventions, social and ecological-economic implications have forced a rethinking the world over. Starting from the statement of the Dublin Conference on Water and Environment in 1992, many experts have emphasised the need for a new paradigm for water management.

Primary inputs of freshwater to India, in the form of rain and snowfall, are strongly influenced by the monsoon and are highly skewed in time and unevenly distributed over the country. Annual average precipitation over India amounts to about 4,000 billion cubic metres. A close examination of smaller areas gives rise to difficulties. The arid areas in Rajasthan receive about 200 mm of annual rainfall, while many areas in Meghalaya annually receive over 11,000 mm of rainfall.

Further, about 80 per cent of this rainfall occurs during two-and-half months, from July to September. This spatial and temporal inequity in the natural distribution of precipitation makes way for some serious challenges in water management in India.

Among the various requirements of water, safe drinking water should get the highest priority. In India, irrigation has received the highest investment and irrigation potential has

increased from about 20 million hectare in the pre-Plan era of the 1950s to about 102 million hectare by the turn of the century. This

300 mn

THE NUMBER OF INDIANS WHO DON'T HAVE ACCESS TO SAFE DRINKING WATER. YET AN AVERAGE OF 1,000 CUBIC METRES IS USED BY EVERY INDIAN PER YEAR.

Diminishing Lifeline

WATER ENGINEERING IN INDIA HAS FOCUSED ON IRRIGATION, HYDRO-POWER GENERATION AND FLOOD CONTROL. THE HIGH PRIORITY BUT LOW VOLUME NEEDS OF DOMESTIC DRINKING WATER HAVE BEEN IGNORED.

Endless thirst: A third of India does not get safe drinking water